

YOC.ai Camera Scoring Model

Current implementation → target seven-metric model → JSON + lightweight AI analysis

1. Current Camera Scoring Model

The current MVP scorer uses five categories: Presence, Framing, Head Control, Posture, and Stability. These are heuristic product scores rather than psychological or clinical measurements.

Category	Current Formula	Weight
Presence	Face Visibility %	20%
Framing	Good Framing % × 0.70 + Centered Face % × 0.30	15%
Head Control	Head Centered % × 0.60 + Away Component × 0.40	25%
Posture	Good Posture % × 0.75 + Body Centered % × 0.25	25%
Stability	Head Stability × 0.55 + Body Stability × 0.45	15%

Current Overall Formula

$$\text{Overall} = \text{Presence} \times 0.20 + \text{Framing} \times 0.15 + \text{Head Control} \times 0.25 + \text{Posture} \times 0.25 + \text{Stability} \times 0.15$$

Example

If Presence = 95, Framing = 90, Head Control = 86, Posture = 89, and Stability = 92:

$$95 \times 0.20 + 90 \times 0.15 + 86 \times 0.25 + 89 \times 0.25 + 92 \times 0.15 = 90.05 \rightarrow 90/100$$

2. Current Metric Details

Metric	How it is currently calculated
Presence	Face Visibility %. Example: 930 visible samples / 1000 face samples = 93.
Framing	70% Good Framing + 30% Centered Face.
Head Control	Head Centered % contributes 60%. Away Component contributes 40%. Away Component = clamp(100 – Looking
Posture	75% Good Posture + 25% Body Centered.
Stability	55% Head Stability + 45% Body Stability.

3. Reliability

Reliability is separate from the mathematical score. The current scorer requires at least 60 face samples, 30 head samples, and 20 pose samples. If these thresholds are not met, the result is marked unreliable rather than automatically reducing the score.

4. Blink Count and Blink Rate

Removed permanently from the product scoring model. Blinking is natural and should not be treated as an interview-performance signal. Eye openness telemetry may remain useful internally, but blink count and blink rate should not appear in the final report or contribute to scoring.

5. Target Final Camera Performance Model

The intended user-facing result contains exactly seven primary metrics:

Final Metric	Meaning
Camera Engagement	How consistently the candidate remained visually engaged with the camera.
Posture	Upright, balanced upper-body posture versus leaning/slouching.
Facial Expressiveness	Natural variation in facial expression during responses.
Head Stability	Controlled/natural head movement versus excessive or distracting movement.
Framing	Face position, distance from camera, and centering.
Visual Quality	Face visibility and overall suitability of the visual input for analysis.
Presence Stability	How consistently the candidate remained visible, trackable, and properly positioned.

Recommended Product Split

Behavior: Camera Engagement, Posture, Facial Expressiveness, Head Stability, Presence Stability.

Setup: Framing, Visual Quality.

Visual Quality should not contribute heavily to the overall performance score. A poor webcam or dark room should not make YOC.ai imply that a candidate is a worse interviewer.

6. Presentation Consistency

Presentation Consistency is internal/session-level telemetry, not an eighth user-facing score.

$$\text{Presentation Consistency} = \text{Good Presentation Samples} / \text{Valid Presentation Samples} \times 100.$$

It should use temporal persistence and smoothing. Brief glances, one noisy MediaPipe frame, blinking, small head movements, or normal facial movement should not immediately count as instability. Sustained or repeated problems can contribute to instability.

7. Important Posture Constraint

Hips should not be required. The posture engine should be based on the upper body and should work with the shoulder region: shoulder visibility, alignment/tilt, upper-body centering, and stability.

8. Missing Pieces Before Final Scoring

Component	Status
Camera Engagement	Needs final formula
Posture	Exists, but must be made shoulder-only
Facial Expressiveness	Needs expression-variation telemetry
Head Stability	Existing head telemetry can support it
Framing	Existing telemetry can support it
Visual Quality	Needs real measurable quality signals
Presence Stability	Needs session-level stability/continuity logic

9. Target Data Flow

Camera → MediaPipe Models → Feature Extraction → Session Aggregation → Presentation Metrics → Seven Final Scores → JSON Report → Lightweight AI Analysis

10. JSON Report Concept

```
{
  "camera_performance": {
    "overall": 87,
    "camera_engagement": 82,
    "posture": 88,
    "facial_expressiveness": 79,
    "head_stability": 86,
    "framing": 92,
    "visual_quality": 89,
    "presence_stability": 95
  },
  "telemetry": {
    "looking_away_percent": 8.4,
    "face_visibility_percent": 97.2,
    "good_framing_percent": 91.5,
    "good_posture_percent": 88.2,
    "presentation_consistency": 94.1
  }
}
```

The AI should receive the summarized structured JSON rather than raw MediaPipe landmarks. The deterministic scoring engine remains responsible for numerical scores; the AI becomes the explanation/feedback layer.

11. Lightweight AI Layer

A lightweight model is sufficient for JSON-to-feedback. Gemini 2.5 Flash-Lite is a strong candidate for this role because the task is structured-input interpretation and feedback generation rather than raw computer-vision reasoning.

- Recommended responsibility split:**
- TypeScript scoring engine → calculates scores deterministically.
 - JSON report → compact session representation.
 - Lite AI model → explains strengths, weaknesses, and 2-3 actionable recommendations.

12. Implementation Order

Step	Work
1	Permanently remove Blink Count / Blink Rate.
2	Make posture genuinely shoulder-only.
3	Add/finalize Presentation Consistency telemetry.
4	Add facial-expression telemetry.
5	Add measurable Visual Quality telemetry.
6	Design and validate the seven final scoring formulas.
7	Test scoring against multiple sessions.
8	Freeze the scoring engine.

Step	Work
9	Produce the clean JSON report.
10	Feed the JSON to a lightweight AI model for feedback.
11	Build the final user-facing camera report UI.

Core principle: measure behavior over time, not isolated frames. Reward consistent good presentation rather than perfect stillness.